

Assignment 1 Memo

Pittsburgh 311: Parks, Trees, Animals, and Public Facilities

1. Corpus Construction

The corpus focuses on Pittsburgh 311 service requests related to four categories: Parks Issues, Tree Issues, Animal Issues, and City Facilities and Infrastructure. The primary data sources were the WPRDC 311 request dataset and the accompanying issue and category codebook.

The corpus contains 224 records covering 28 official Pittsburgh 311 issue types. For each issue, I created eight records: six short resident-style complaint examples and two structured operational records. The text examples were designed to include clear, ambiguous, incomplete, typo-heavy, informal, and short complaint styles. The structured records contain request volume and resolution-time statistics derived from the historical 311 data.

Each record includes the official request type ID, issue, category, and responsible department in its metadata. Synthetic complaint examples are explicitly marked as synthetic, while operational records derived from the WPRDC dataset are marked as non-synthetic.

2. Corpus Characterization

The final corpus contains 224 records, including 168 text records and 56 table records, giving a tabular share of 25%.

All records come from the WPRDC domain. This source concentration is intentional because the project focuses specifically on Pittsburgh 311 service requests and uses the official 311 request data and codebook as the authoritative source.

The corpus contains no near-duplicate pairs. The median document length is 54 characters, reflecting the intentionally short style of many resident complaint examples. Because the corpus only covers four Pittsburgh 311 categories, it cannot represent the full range of municipal service requests.

3. Human vs. Phi Evaluation

A random sample of 25 records was manually labeled and then evaluated with Phi using the original extraction prompt. The baseline produced no records with all four fields correct. Field-level accuracy was 16% for `request_type_id`, 24% for `issue`, 80% for `category`, and 0% for `department`.

Although the model usually identified the broad category correctly, it often paraphrased official issue names or generated non-official department labels. For example, it produced labels such as "Unsafe tree" instead of "Tree Removal" and "Parks and Recreation" instead of the official responsible department.

4. Error Analysis

The main baseline errors came from exact-label matching rather than broad topic recognition. Phi often understood the general meaning of a complaint but returned a paraphrased issue name or a non-official

department name. It also struggled to infer official `request_type_id` values from complaint text when the numeric ID was not explicitly present.

Some records were also genuinely ambiguous. For example, a complaint describing an unsafe tree could reasonably resemble either Tree Removal or a fallen-tree issue. These cases show that semantic similarity alone is not enough for reliable routing when the system must output exact official Pittsburgh 311 labels.

5. Recovery Prompt Experiment

To address the recurring exact-label errors, I revised the prompt to include the official Pittsburgh 311 mapping and instructed the model to first choose a `request_type_id` and then copy the issue, category, and department from the same mapping row.

This change substantially improved performance. The number of fully correct records increased from 0 to 15 out of 25. Field-level accuracy improved from 16% to 96% for `request_type_id`, from 24% to 68% for issue, from 80% to 88% for category, and from 0% to 84% for department.

The recovery experiment shows that the model performed much better when the official label space and relationships between fields were explicitly provided. However, some errors remained on ambiguous complaints and cases where Phi still paraphrased the issue label instead of copying the exact official value.

6. Cluster Analysis

The corpus was grouped into six clusters based on text similarity. The resulting silhouette score was 0.120, suggesting that the clusters overlap substantially rather than forming sharply separated groups.

The clusters were interpreted as: Tree Maintenance and Hazards; Animal Health and Cleanup Issues; Pet and Animal Service Requests; Access Obstructions and Blocked Infrastructure; Public Facility Maintenance Issues; and Park Maintenance and Cleanliness Issues.

These clusters generally reflect meaningful operational themes in the corpus. However, some overlap is expected because short 311 complaints often use similar language such as "needs attention," "problem," or "blocking." In addition, table records with empty `raw_text` can weaken the separation between clusters.

On the manually labeled sample, Phi had 0% all-fields-correct accuracy in every cluster under the original prompt. Therefore, no cluster showed clearly stronger baseline performance. This reflects the broader exact-label problem: even when Phi recognized the general topic, errors in `request_type_id`, issue, or department caused the entire record to be counted as incorrect.

7. Limitations

The corpus has several limitations. First, all records come from a single WPRDC domain, so source diversity is limited. Second, the text records are synthetic resident-style examples rather than original free-text 311 complaints, which may not fully capture the language, noise, and ambiguity of real submissions.

The corpus also covers only four Pittsburgh 311 categories, so the findings do not generalize to the full 311 taxonomy. In addition, many records are very short, and some tabular records have empty `raw_text`,

which can make clustering less informative. Finally, exact-label evaluation is strict, so semantically reasonable paraphrases are still counted as incorrect.

8. Conclusion

This project shows that Phi can recognize the broad topic of Pittsburgh 311 complaints, but reliable routing requires access to the official taxonomy and exact label mappings. The original prompt performed poorly on exact request IDs, issue names, and departments, even when its semantic interpretation was reasonable.

Providing the official mapping and requiring consistent labels from the same mapping row substantially improved extraction accuracy. The remaining errors were concentrated in ambiguous or very short complaints, suggesting that a practical 311 routing system should combine structured taxonomy knowledge with clarification questions when the user's description is insufficient.